



Development of a millimeter-wave sensor system for versatile sensing applications

Yuheng Li

1569107, yuhengl8@student.unimelb.edu.au

Dongfang Lyu

1444967, dongfang.lyu@student.unimelb.edu.au

Haohan Xu

1512074, haohan.xu3@student.unimelb.edu.au

Executive Summary: This report details the development of an open, end-to-end millimeter-wave (mmWave) radar sensing platform designed for versatile sensing applications. Utilizing a 60 GHz pulsed coherent radar sensor, the project addresses the growing demand for compact, low-power object detection and distance measurement solutions across various domains, including autonomous driving, smart parking, and healthcare monitoring. The primary objective was to engineer a complete full-stack architecture, encompassing an STM32-based custom printed circuit board, robust embedded firmware, and a Bluetooth Low Energy (BLE) companion mobile application. The extent to which this report establishes a novel view lies in its delivery of an integrated, highly adaptable hardware-software ecosystem that achieves sub-millimeter measurement accuracy while optimizing power consumption to less than 1 μA in its idle state. Specifically, the system successfully implements and validates two advanced operational modes: a high-frequency vibration monitoring mode capable of detecting microscopic displacements, and a contactless vital signs monitoring mode that extracts real-time breathing and heart rates through sophisticated phase-unwrapping and spectral analysis algorithms. Furthermore, the foundational algorithm and state machine for a third application, fall detection, have been designed and partially integrated, establishing a clear pathway for future work. By synthesizing complex digital signal processing with real-time mobile visualization, this research provides a comprehensive and scalable foundation for rapid deployment in diverse industrial and clinical sensing scenarios.

1 Introduction

1.1 General overview of the project topic and why it matters to society.

The ability to accurately monitor both environmental factors and human health metrics without physical intrusion is critical across numerous domains, ranging from predictive maintenance in industrial machinery to non-wearable health tracking and elderly care. Traditional sensors—such as contact-based accelerometers, wearable heart monitors, or camera-based fall detectors—can be difficult to deploy, uncomfortable for continuous wear, or raise privacy concerns. This project introduces a complete, open radar sensing platform utilizing the Acconeer A121, a 60.5 GHz pulsed coherent radar sensor characterized by its ultra-compact 29 mm² footprint and exceptionally low power consumption. By operating in the mmWave band, the system leverages coherent radar technology to detect sub-millimetre displacements and subtle phase shifts completely contact-free.

1.2 Overall aim(s) of the project.

The overall aim of this project is to design and implement a cohesive, end-to-end sensing solution encompassing a custom hardware PCB, STM32WBA64 microcontroller firmware, Bluetooth Low Energy (BLE) connectivity, and a mobile application interface. Specifically, the project focuses on developing three core operational modes on this shared architecture: high-precision vibration monitoring for frequency and displacement analysis, non-contact vital signs tracking for heart and breathing rates, and a state-machine-driven fall detection algorithm. Together, these modes demonstrate a robust, real-time reference design for advanced remote sensing applications.

2 Literature Review

2.1 Existing research on our topic

Millimeter-wave (mmWave) pulsed coherent radar (PCR) has emerged as a highly promising technology for portable, non-contact physiological monitoring due to its exceptional capability to detect extremely small body-surface displacements. Unlike conventional continuous-wave architectures, PCR operates by transmitting ultra-short, phase-coherent pulses with an extremely low duty cycle. This unique signalling mechanism enables microwatt-level power consumption and high peak energy efficiency, making it exceptionally suited for battery-powered, portable edge devices. Recent advances in high-frequency PCR hardware demonstrate that micrometer-scale surface motion can be accurately captured under real-world conditions, making mmWave PCR highly practical for continuous heartbeat monitoring, respiration sensing, and fall detection. Compared with wearable sensors and camera-based systems, mmWave PCR offers identical advantages in privacy preservation and non-invasive operation while strictly complying with the stringent power and size constraints of smart-health applications.

Instead of using beat frequency analysis typical of continuous-wave radar, PCR estimates target distance by evaluating the time-of-flight (ToF) of reflected pulses across discrete range gates. Within a specific range gate, the pulse-to-pulse phase coherence enables highly accurate displacement estimation at the micrometer scale (Ikram et al., 2018). This phase-based sensing capability is crucial for healthcare applications, as physiological activities such as heartbeat and respiration produce minute chest-wall movements ranging from several tens to hundreds of micrometers. To achieve single-digit micrometer precision and reduce hardware-induced jitter or distortion within the range gates, researchers have proposed phase-evaluation and intra-gate calibration algorithms that compensate for pulse timing errors and distance-dependent nonlinearities. Such calibration methods have been demonstrated to improve local ranging accuracy from hundreds of micrometers to single-digit micrometers across meter-scale detection ranges.

Additional signal-processing approaches further enhance PCR sensitivity and robustness. Dual-phase-term tracking and high-rate coherent pulse integration methods suppress clock instabilities and phase ambiguity, enabling micro-displacement estimation with high accuracy even under strict hardware resource constraints. These studies suggest that advanced phase-processing algorithms are critical for extracting weak physiological signals from noisy, discrete radar measurements in real-time monitoring environments.

Several studies have demonstrated that high-frequency mmWave PCR systems can achieve displacement variances approaching the theoretical Cramér–Rao lower bound. By leveraging high-rate coherent pulse integration, researchers achieved displacement variances below 10 μm (Ikram et al., 2018). Miniaturized PCR sensors operating at 60 GHz and advanced SiGe sub-terahertz systems further improve phase sensitivity to the single-digit micrometer level over short-range monitoring scenarios.

Comparative studies between different carrier frequencies show that higher operating frequencies provide superior phase sensitivity and repeatability. This improved sensitivity is particularly beneficial for detecting weak heartbeat-induced chest vibrations that may otherwise be obscured by respiratory motion or environmental interference. However, higher operating frequencies also introduce challenges including increased propagation loss, hardware complexity, and stricter calibration requirements, highlighting the trade-off between sensing precision and practical deployment cost in portable hardware design.

For practical healthcare applications, robustness against clutter, involuntary body movement, and environmental interference is essential. Several studies emphasize algorithms designed to suppress disturbing targets and micro-vibrational interference while maintaining high pulse repetition rates for real-time monitoring. Modern coherent pulse systems have also demonstrated the ability to maintain micrometer-level accuracy for moving targets, achieving displacement accuracy below 30 μm for velocities up to 3 m/s. These capabilities are highly relevant to fall detection and continuous motion monitoring, where large-scale body movement introduces significant phase distortion.

Despite these advances, separating weak heartbeat signals from larger body motion remains a major technical challenge. Physiological monitoring environments are considerably less controlled than laboratory ranging experiments, and radar reflections from furniture, walls, or multiple subjects can introduce severe multipath interference within the target's range gate. This challenge is further compounded during macro-movements like falls, which can completely saturate the receiver phase and mask the subtle micro-displacements required for vital sign tracking. Consequently, reliable clutter suppression and motion-artifact removal remain active research areas in non-contact healthcare sensing.

Although wide physical bandwidth improves radar resolution, practical portable systems are often limited by hardware complexity and power consumption. To address these limitations within the PCR paradigm, researchers have developed digital signal-processing techniques that enhance effective range and phase resolution without expanding physical hardware footprints. Adaptive signal-expansion and padding techniques improve target detection performance using existing hardware resources. Similarly, compressed sensing, sparse sampling, and optimized range-gate interpolation strategies enable high range and velocity resolution while using only a fraction of the conventional power and computational budget. These methods are particularly valuable for portable medical-monitoring systems, where low power consumption, compact hardware size, and real-time processing capability are critical design requirements.

2.2 Research gap(s) our project will address

Overall, the literature demonstrates that mmWave PCR possesses the sensitivity, low-power profile, and resolution required for non-contact physiological monitoring. Existing research has shown strong performance in displacement sensing, motion tracking, and environmental robustness using advanced phase-processing and calibration algorithms. However, several important research gaps remain.

First, many existing high-precision radar studies focus primarily on industrial distance-gauging rather than healthcare-oriented monitoring applications.

Second, current research often investigates heartbeat detection, respiration sensing, or fall detection independently, while fewer systems integrate all three functions into a single, unified portable real-time framework.

Third, reliable extraction of weak heartbeat signals in the presence of large body motion, clutter, and multipath reflections remains insufficiently addressed in realistic indoor environments, particularly under the constraints of low-power, discrete-range-gated PCR hardware.

Therefore, this project aims to develop a robust, computationally efficient mmWave PCR signal-processing framework suitable for portable hardware deployment, capable of simultaneously performing heartbeat monitoring, respiratory detection, and fall recognition using non-contact sensing. By combining discrete range-gating techniques with advanced motion-separation and filtering algorithms, the project seeks to improve the practicality and reliability of continuous healthcare monitoring for elderly-care and smart-health applications.

3 Methodology/Methods

3.1 Available Methodologies

To address the research gaps regarding sub-millimeter sensing accuracy and ultra-low power consumption, several radio-frequency (RF) methodologies were evaluated. The primary available radar architectures include:

1. **Frequency Modulated Continuous Wave (FMCW):** Offers excellent range and velocity resolution by continuously sweeping a frequency band. However, its continuous active state results in high power consumption, rendering it unsuitable for long-term battery operation.
2. **Ultra-Wideband (UWB) Impulse Radar:** Emits extremely short pulses to measure distance via time-of-flight. While highly power-efficient, traditional UWB often struggles to extract precise phase information required for micro-vibration analysis.

3. **Pulsed Coherent Radar (PCR):** Combines the power efficiency of pulsed systems with the high precision of coherent receivers. By transmitting discrete pulses and analyzing the phase of the received echo, PCR can achieve high-resolution measurements with significant duty-cycling capabilities.
4. **Continuous-wave (CW) Doppler radar.** A CW radar transmits a single-frequency carrier and measures the Doppler frequency shift of the reflected signal, which is proportional to the radial velocity of the target. Although simple to implement, CW radar produces no range information: all reflectors within the antenna beam contribute to the output, making it highly susceptible to clutter from moving objects in the environment. Furthermore, the Doppler spectrum of respiration contains harmonics that overlap with the cardiac frequency band, complicating heart-rate extraction.

3.1.1 Physiological Sign Monitoring

Several sensing modalities exist for non-obtrusive physiological monitoring of breathing rate and heart rate.

Contact-based sensors. Conventional clinical instruments — including electrocardiography (ECG) electrodes, pulse oximeters, and respiratory chest straps — provide high-accuracy measurements but require direct skin contact or active cooperation from the subject. For continuous, unattended monitoring of elderly individuals, long-term wearability presents a significant compliance challenge; devices may be forgotten, removed during sleep, or improperly fitted.

Camera-based methods. Remote photoplethysmography (rPPG) extracts cardiac signals from subtle colour changes in facial skin captured by a standard RGB camera. Thermal imaging can additionally detect respiratory airflow patterns. While contact-free, both approaches are sensitive to ambient lighting conditions, require a clear and unobstructed line of sight to the subject's face or skin, and raise privacy concerns in domestic environments.

3.1.2 Fall Detection

Wearable inertial sensors. Accelerometers and gyroscopes integrated into wristbands, pendants, or smart watches can reliably distinguish fall dynamics from normal activities of daily living. However, their effectiveness is entirely dependent on continuous wearing; studies consistently report low long-term compliance among elderly users, with devices frequently left uncharged or removed at night — precisely when fall risk is elevated.

Camera and computer vision. Vision-based systems analyse video streams to detect posture transitions associated with falls. Deep learning approaches have demonstrated competitive accuracy in controlled settings, but real-world deployment is constrained by lighting requirements, occlusion from furniture, and significant privacy concerns that may deter installation in bedrooms or bathrooms.

Floor pressure and vibration sensors. Pressure-sensitive mats or floor-embedded vibration sensors can detect the impact of a fall through the characteristic force signature. Their coverage is inherently limited to instrumented areas, and installation requires modification of the living environment.

Radar-based detection. Radar sensors detect falls non-invasively without requiring contact with the subject or visible illumination. By monitoring both the energy of rapid motion (impact phase) and the subsequent displacement of the subject from their pre-fall position, radar can confirm falls with low latency while rejecting false alarms from normal activities.

3.2 Chosen Methodology

3.2.1 Radar Sensor

The Acconeer A121 60 GHz pulse-coherent radar was selected as the primary sensing element for this system. The choice of radar over contact-based or camera-based alternatives was motivated by three

requirements: the system must operate continuously without subject cooperation, it must function in low-light and occluded conditions, and it must respect user privacy. Radar satisfies all three criteria.

Among the available radar architectures, the pulsed coherent approach of the A121 was preferred over CW Doppler and FMCW radar for physiological monitoring. As described by the phase-displacement relationship in $\Delta d = \frac{\lambda}{4\pi} \Delta\phi$, the IQ output provides direct access to phase information, allowing small chest-wall displacements to be estimated from phase variations. For a 60 GHz radar with a wavelength of approximately 5 mm, this relationship provides a displacement sensitivity of about 0.40 mm/rad, which is sufficient for detecting cardiac-induced chest-wall motion on the order of 0.1–0.5 mm. In addition, unlike FMCW radar, the pulsed coherent architecture does not require continuous frequency sweeping, reducing the phase-noise challenges associated with low-amplitude motion sensing. A single A121 sensor can therefore support both vital-sign monitoring and fall detection, avoiding the complexity of a multi-sensor fusion system.

The 60 GHz operating frequency was also advantageous from a hardware perspective: the short free-space wavelength ($\lambda_0 = 5$ mm) allows a compact antenna aperture while achieving high phase sensitivity, and the millimetre-wave band penetrates clothing without raising ionising radiation safety concerns.

3.2.2 Wireless Data Transmission

Bluetooth Low Energy (BLE) was selected for communication between the sensor node and a mobile application. Compared with Wi-Fi, BLE offers lower power consumption and does not require connection to a local network infrastructure, enabling standalone operation. Compared with ZigBee or proprietary sub-GHz protocols, BLE provides native compatibility with consumer smartphones without additional gateway hardware. The STM32WBA64 microcontroller integrates a BLE 5.4 radio alongside the Cortex-M33 application processor, allowing radar processing and wireless transmission to be handled within a single chip without an external radio module.

3.2.3 Lens Design

The A121 sensor module produces a naturally narrow, high-directivity beam at 60 GHz. For physiological monitoring, a narrow beam risks missing the subject's chest wall when the subject shifts position slightly, degrading measurement continuity. A Fresnel Zone Plate dielectric lens was therefore designed and 3D-printed using thermoplastic material to widen the beam pattern and provide broader angular coverage of the monitoring zone. The lens geometry was derived using Acconeer's official lens design tool, which computes the stepped dielectric ring profile required to produce a specified beam divergence at the sensor operating frequency. The use of 3D printing enabled rapid iteration of the lens geometry and straightforward mechanical integration with the radar housing enclosure.

3.2.4 Fall Detection Algorithm

Stillness-based fall confirmation — requiring the subject to remain motionless for an extended period after a suspected impact — was considered but rejected. Conscious elderly subjects who have fallen frequently continue to move, struggle to rise, or call for help; a stillness criterion would therefore fail to confirm genuine falls in this population. Instead, the implemented algorithm confirms a fall by verifying that the subject's estimated distance from the sensor 5 has shifted by at least $\tau_d = 0.20$ m from the pre-impact position and remains displaced for $t_{\text{confirm}} = 5$ s. This distance-based criterion is satisfied regardless of whether the subject is stationary or moving at the fallen position, making it robust to conscious patients.

3.3 Hardware Design Methodology and Implementation

3.3.1 Core Hardware Design Philosophy: Risk Mitigation and Dual-Constraint Alignment

Given the vast array of electronic components available in the current market, the hardware design was governed by a highly pragmatic, risk-driven philosophy. In hardware engineering, unlike software development, design flaws are notoriously difficult to debug and require costly, time-consuming PCB respins (re-designing, re-manufacturing, and re-assembling the physical board). Therefore, the design logic prioritized minimizing the probability of failure while strictly satisfying two distinct categories of requirements.

- **The first category encompasses manufacturer-imposed silicon constraints:** This includes the strict power sequencing, decoupling, and RF layout requirements mandated by the IC manufacturers (specifically Acconeer and STMicroelectronics) to ensure signal integrity and hardware stability.
- **The second category comprises the project-specific constraints:** Namely, achieving ultra-low-power and battery-operated deployment.

3.3.2 Iterative design methodology

The hardware development followed a highly iterative, non-linear design methodology, which is essential for a custom, from-scratch implementation. Rather than strictly adhering to a sequential pipeline, the design evolved through continuous feedback loops based on a macro-level system architecture. Before initiating the custom printed circuit board (PCB) routing, official evaluation boards and hardware design guidelines from both Acconeer and STMicroelectronics were rigorously studied. Crucially, for the foundational peripheral sub-systems—such as the MCU's decoupling capacitor networks, crystal oscillators, RF, and boot configurations—the custom PCB strictly adopted the manufacturers' proven reference layouts. In professional hardware engineering, deviating from official reference designs for these standard circuits introduces unnecessary parasitic variables and severely compromises baseline stability. Therefore, directly leveraging these verified topologies was a calculated decision, ensuring a robust processing foundation while focusing developmental efforts on the novel integration of the mmWave sensor. This prototyping phase was critical for reverse-engineering these reference designs and validating the intertwined hardware-software architecture—such as porting the Acconeer RSS SDK to the Cortex-M33 architecture and defining precise SPI topologies. The finalized macro-level hardware architecture is illustrated in **Figure 1**.

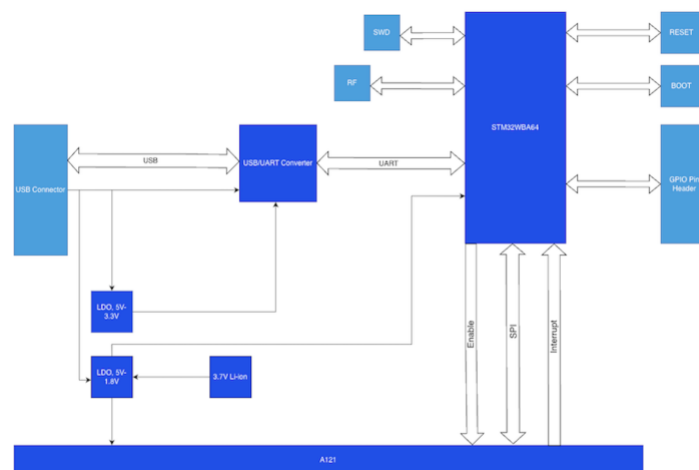


Figure 1: High level hardware architecture block diagram

3.3.3 Component selection and risk-driven architectural refinement

The component selection phase was deeply intertwined with the architectural refinement, requiring a delicate balance between processing capabilities and development risk mitigation. Initially, the ESP32 microcontroller was considered due to its widespread popularity. However, a critical evaluation of its official hardware integration resources revealed a lack of the granular, rigorous documentation required for a first-time, custom PCB design. This sparsity in official reference material significantly elevated the probability of critical hardware design errors. In contrast, the STMicroelectronics ecosystem provided exhaustive, professional-grade hardware design guidelines and reference schematics. Consequently, the STM32WBA64 was selected; the completeness and rigor of its official documentation were deemed the paramount factor in minimizing hardware iterations and mitigating developmental risks. Furthermore, as an architectural bonus, its 100 MHz Arm Cortex-M33 core provided superior digital signal processing (DSP) headroom for edge-computing, and its native support for the UTIL_SEQ scheduler allowed seamless interleaving with the Acconeer A121's strict SPI interrupt timing.

3.3.4 mmWave Sensor Integration and Power Architecture

The sensing capability is driven by the Acconeer A121, a 60.5 GHz pulsed coherent radar sensor. To address the power constraints typical of radar systems and enable untethered deployment, a hard-power gating methodology was implemented. The STM32 actively toggles the sensor's `ENABLE` pin, completely cutting power to the radar between discrete measurement sweeps. To support this, the platform is powered by a 3.7 V single-cell lithium-ion battery, managed by a Texas Instruments BQ25176K linear charger IC. Crucially, this specific charger was selected primarily for its highly simplified, low-pin-count architecture. In the context of a first-time custom PCB design, minimizing the component pin count drastically reduces trace routing complexity and physical soldering difficulty. This pragmatic choice directly aligns with the project's risk-driven design philosophy by minimizing potential points of failure. Furthermore, the custom PCB's power tree was bifurcated to protect RF signal integrity: a 1.8 V Low-Dropout (LDO) regulator explicitly supplies the MCU and A121 domains, while a separate 3.3 V LDO isolates the onboard CP2105 USB-UART bridge. Ultimately, this architecture achieves an overall idle current draw of less than 1 μA . The detailed power distribution network, highlighting the bifurcation between the 1.8V domain and the 3.3V domain, is shown in **Figure 2**.

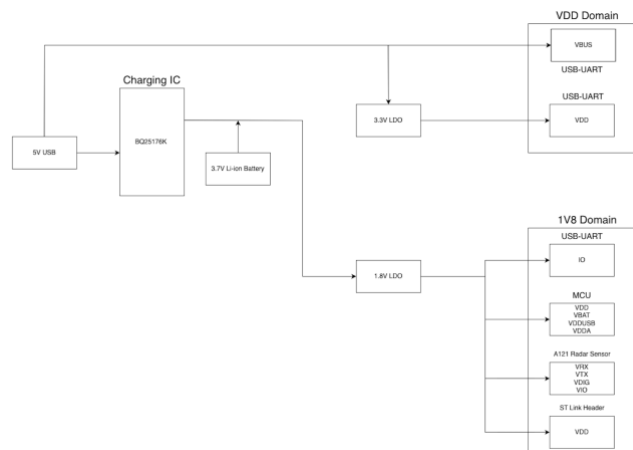


Figure 2: Power Tree

3.3.5 Printed circuit board stack-up and layout

Given the extreme 60.5 GHz operating frequency of the Acconeer A121 sensor and the 2.4 GHz BLE radio, the physical PCB layout required rigorous attention to Signal Integrity (SI) and Power Integrity (PI). To accommodate these demands within an ultra-compact footprint, a 4-layer PCB stack-up was engineered. Specifically, Layer 2 was designated as a dedicated 1.8 V power plane. Since all active

components—including the STM32's power domains and the A121 radar—operate exclusively on 1.8 V, this dedicated plane significantly reduces the Power Delivery Network (PDN) impedance, ensuring a highly stable voltage supply free from resistive drops. Furthermore, an unbroken internal ground (GND) plane was implemented. This pure ground layer is critical; it provides a continuous, low-impedance return path for the high-frequency signals, minimizing loop inductance and suppressing electromagnetic interference (EMI).

In terms of overall component placement, the design strictly adhered to manufacturer hardware integration guidelines to ensure baseline stability. Specifically, the PCB layouts for the STM32WBA64 microcontroller, the CP2105 USB-UART bridge, and the BQ25176K charging IC replicated their respective official reference designs. Adopting these proven topologies minimizes parasitic variables and guarantees reliable power delivery and communication.

Regarding the A121 sensor, it was isolated on the bottom layer of the PCB, physically distancing it from the active components to prevent near-field interference. The official datasheet strongly recommends avoiding any trace routing on the sensor's immediate layer to eliminate parasitic coupling. However, the spatial constraints of the project's objectives necessitated a calculated engineering compromise. Due to routing bottlenecks, some traces had to be routed on the sensor layer. To actively mitigate this risk, these traces were strategically routed to bypass the sensor's designated RF keep-out zones. This pragmatic trade-off successfully achieved the required ultra-compact form factor without significantly degrading the radar's directional radiation pattern or micrometer-level phase accuracy.

3.3.6 Physical Implementation and Validation

The transition from a schematic to a functional physical prototype presented significant engineering challenges that could only be resolved through empirical validation. The PCB layout required meticulous attention to high-frequency design principles, particularly concerning impedance matching and the strategic placement of decoupling capacitors to maintain power integrity for the 60 GHz A121 sensor. Furthermore, the physical assembly and soldering of the ultra-compact components required extreme precision. The true efficacy of the hardware design could only be verified post-assembly during the hardware "bring-up" phase. This hands-on validation process involved sequentially testing the power rails for stability, verifying the integrity of the high-speed SPI communication, and ultimately evaluating the radar's RF propagation. This empirical debugging phase was crucial; it exposed physical realities—such as power supply noise and parasitic capacitance—that software simulation tools often obscure, confirming that the physical board successfully met the target power profile and sensing capabilities. The final PCB layout, demonstrating the highly compact component placement and RF antenna routing, is shown in **Figure 3**.

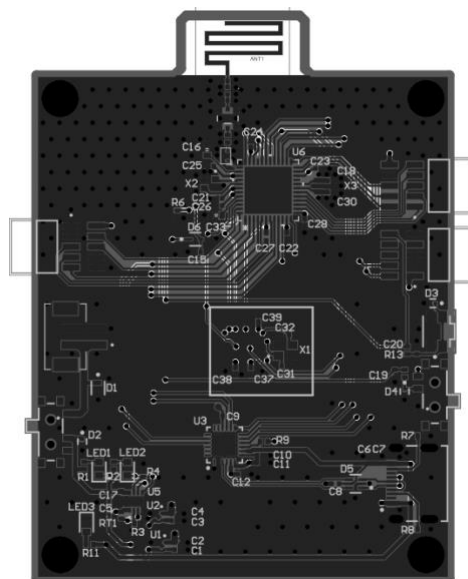


Figure 3: Printed Circuit Board (PCB) layout design

3.3.7 Hardware debugging and PCB rework

Despite rigorous iterative reviews during the schematic capture phase, the physical bring-up of the V1.0 prototype revealed a practical reality of custom hardware engineering. Upon the initial soldering assembly and connecting the device via USB, the host PC failed to enumerate the CP2105 USB-UART bridge. A systematic hardware debugging process was immediately initiated. After ruling out localized soldering defects and verifying the LDO voltage levels, the fault was ultimately traced back to an oversight in the original schematic: the USB data differential pair (D+ and D-) had been inadvertently reversed. To rectify this without enduring the severe time and financial costs of a PCB respin, precision physical rework was executed. The erroneous copper traces were manually cut off, and micro-jumper wires (fly-wires) were soldered to cross-connect and restore the correct differential polarity. While this debugging process was time-consuming, it highlighted a significant engineering triumph: despite this localized interface error, the vastly more complex subsystems—namely the bifurcated LDO power delivery network, the 60 GHz mmWave radio and the BLE radio—functioned absolute data reliability. This practical troubleshooting experience not only restored full system functionality but also rigorously validated the resilience of the core hardware architecture.

3.4 Firmware architecture and system integration

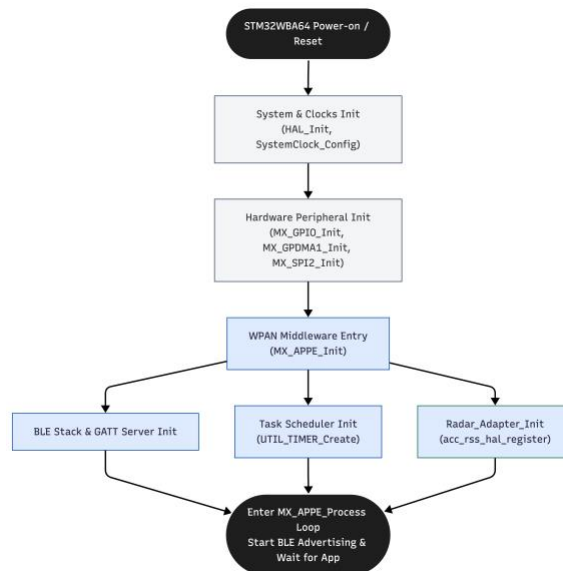


Figure 4: STM32 firmware boot sequence executing within `main.c`

3.4.1 Microcontroller initialization and peripheral configuration

The foundation of the firmware architecture was established using the STM32CubeMX environment to generate the Hardware Abstraction Layer (HAL). Following the physical hardware validation, the immediate firmware objective was to configure the MCU's internal routing and peripheral interfaces to match the custom PCB layout. This involved precisely mapping the General-Purpose Input/Output (GPIO) pins for the A121's dedicated `ENABLE` and `INTERRUPT` lines. Furthermore, the communication buses were initialized: a high-speed Serial Peripheral Interface (SPI) was configured for raw radar data acquisition, and a Universal Asynchronous Receiver-Transmitter (UART) interface was established for real-time debugging. Crucially, the system's clock tree was meticulously configured. Achieving tight synchronization is vital for pulsed coherent radar; therefore, the core clock and peripheral bus frequencies were optimized to ensure that the SPI transactions and the BLE radio subsystem could operate concurrently without timing violations or data bottlenecks.

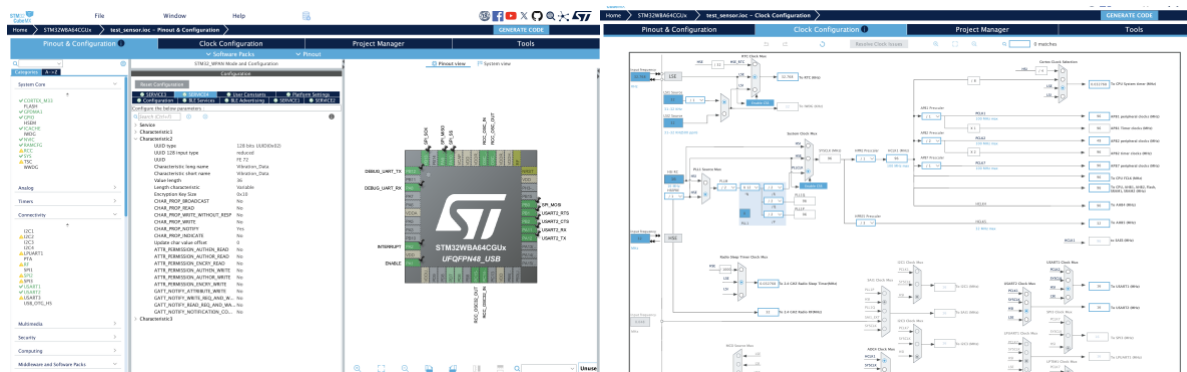


Figure 5: Microcontroller initialization and peripheral configuration

3.4.2 Build system integration and SDK deployment

A critical phase in the firmware development was the integration of the Acconeer Radar System Software (RSS) SDK. While Acconeer officially provides an SDK version compatible with the Arm Cortex-M33 architecture, incorporating this external resource into the custom STM32 environment presented a distinct build-system challenge. Rather than requiring extensive modifications to the underlying C source code, the primary engineering effort focused on toolchain configuration and build automation. This involved meticulously following official integration guidelines to manually author and configure the `CMakeLists.txt` files tailored to the custom directory structure. This build system configuration was essential to correctly compile the SDK into a static library and establish the necessary linker dependencies. By successfully bridging the Acconeer API with the STM32's hardware abstraction layer (HAL)—specifically linking the SPI read/write routines and the NVIC interrupt handlers—through this custom CMake build process, the complex radar signal processing functions were seamlessly deployed to execute natively on the edge node.

3.4.3 Firmware-level hardware validation (Assembly Test)

Before deploying any complex signal processing algorithms, the integrity of the hardware-software boundary had to be empirically verified. To achieve this, an automated "Assembly Test" was compiled and executed. This specialized diagnostic firmware bypasses high-level logic to directly interrogate the physical hardware connections. The assembly test systematically verified three critical domains: first, it validated the power-gating logic by toggling the `ENABLE` pin and monitoring the sensor's boot sequence; second, it executed a loopback-style test over the SPI bus to confirm that the MCU could successfully read the A121's internal chip ID, verifying bi-directional data integrity; third, it triggered dummy radar sweeps to ensure the physical `INTERRUPT` line was correctly signalling the MCU upon frame completion. Passing this rigorous assembly test provided definitive proof that the custom PCB and the ported firmware stack were communicating flawlessly, establishing a stable foundation for the subsequent vibration detection pipeline.

```

Acconeer software version a121-v1.12.0
00:00:00.006 (I) (assembly_test_a121) Basic read: Test
Bring-up: 'Basic read test' [PASS]
00:00:00.019 (I) (assembly_test_a121) Communication test: Test
00:00:01.225 (I) (assembly_test_a121) Communication test: Effective bitrate = 2807kbps
Bring-up: 'Communication test' [PASS]
00:00:01.242 (I) (assembly_test_a121) Enable pin test: Setup
00:00:01.253 (I) (assembly_test_a121) Enable pin test: Test
Bring-up: 'Enable pin test' [PASS]
00:00:01.267 (I) (assembly_test_a121) Interrupt test: Reset
00:00:01.278 (I) (assembly_test_a121) Interrupt test: Setup
00:00:01.292 (I) (assembly_test_a121) Interrupt test: Interrupt latency[0]: 28us
00:00:01.300 (I) (assembly_test_a121) Interrupt test: Interrupt latency[1]: 28us
00:00:01.308 (I) (assembly_test_a121) Interrupt test: Interrupt latency[2]: 28us
00:00:01.316 (I) (assembly_test_a121) Interrupt test: Interrupt latency[3]: 28us
00:00:01.324 (I) (assembly_test_a121) Interrupt test: Interrupt latency[4]: 28us
00:00:01.331 (I) (assembly_test_a121) Interrupt test: Interrupt latency[5]: 28us
Bring-up: 'Interrupt test' [PASS]
00:00:01.348 (I) (assembly_test_a121) Clock and Supply test: Reset
00:00:01.359 (I) (assembly_test_a121) Clock and Supply test: Setup
Bring-up: 'Clock and Supply test' [PASS]
00:00:01.389 (I) (assembly_test_a121) Sensor calibrate test: Reset
00:00:01.400 (I) (assembly_test_a121) Sensor calibrate test: Setup
Bring-up: 'Sensor calibration test' [PASS]
Bring-up: All tests passed
Application finished OK

```

Figure 6: A121 Assembly Test

To ensure codebase integrity and facilitate systematic debugging throughout the hardware-software integration phase, Git version control was strictly enforced. The complete project repository is available at https://github.com/yuhengli001/test_sensor.git.

3.4.4 Bluetooth Low Energy (BLE) stack initialization and custom GATT logic

With the physical sensing layer validated, the firmware architecture transitioned to establishing the wireless communication backbone. To ensure protocol compliance and stability, the foundational initialization of the STM32WBA64’s integrated BLE 5.4 radio was streamlined using STM32CubeMX, which generated the boilerplate WPAN middleware. However, the critical application-layer logic—specifically bridging the BLE stack with the radar's data pipeline—required extensive custom firmware development. The device was configured to operate as a Generic Attribute Profile (GATT) server. Custom C code was authored to handle the complex callback functions necessary for bi-directional communication: asynchronously parsing incoming control commands (e.g., adjusting radar measurement ranges) from the host App, and packing high-frequency radar data arrays into standardized BLE Notifications for transmission.

Integrating this continuous data transmission introduced severe real-time timing constraints. BLE connection events require strict microsecond-level precision and must not be stalled by the MCU's other operations. Conversely, the A121 radar requires intensive SPI polling. To resolve this concurrency challenge, the firmware leveraged the STMicroelectronics `UTIL_SEQ` cooperative task sequencer. By mapping the radar data acquisition routines and the custom BLE Tx/Rx event handlers as separate, non-blocking tasks within the `UTIL_SEQ` framework, the architecture achieved seamless execution interleaving. The MCU could swiftly service a BLE radio event, yield control, and immediately resume radar data processing.

To standardize the bi-directional communication between the STM32 edge node and the host application, a custom GATT (Generic Attribute Profile) service was architected. This interface was explicitly designed to separate control plane data (e.g., radar configuration commands) from data plane data (e.g., high-frequency continuous radar payloads). The specific custom Universally Unique Identifiers (UUIDs) and their corresponding access properties were defined as follows:

Table 1: GATT Service Profile

Service	UUID	Purpose
Control Service	0000FE40-CC7A-482A-984A-7F2ED5B3E58F	Mode selection, start/stop, sensor status
Vibration Service	0000FE70-CC7A-482A-984A-7F2ED5B3E58F	Vibration config write, vibration data notify
Vital Sign Service	0000FE50-CC7A-482A-984A-7F2ED5B3E58F	Vital sign data notify

3.5 Algorithms

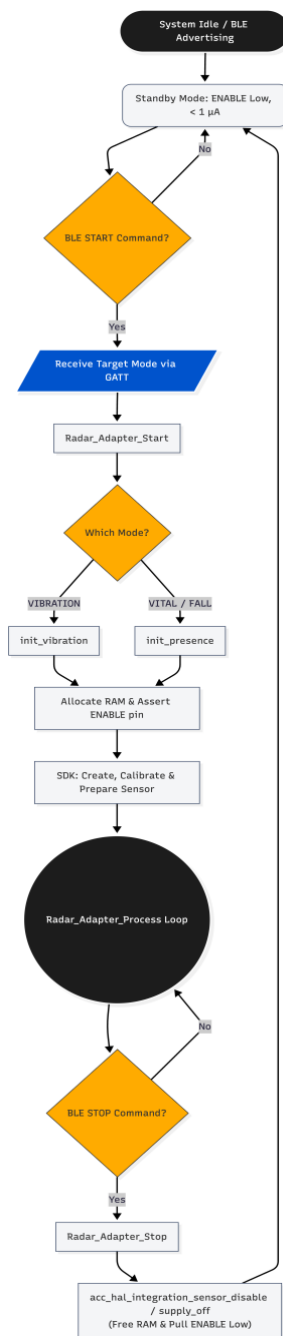


Figure 6. Sensor wake-up, preparation, and measurement state machine within radar_adapter.c.

3.5.1 Sensor preparation and boot sequence

To strictly adhere to the ultra-low-power and compact design objectives, the A121 mmWave sensor is not active continuously. Instead, the firmware implements a rigorously decoupled, two-stage initialization architecture orchestrated by a central firmware abstraction layer (`radar_adapter.c`). During the initial MCU boot sequence, this adapter manages the first stage by calling `Radar_Adapter_Init()` to register the Hardware Abstraction Layer (HAL) with the Acconeer SDK via `acc_rss_hal_register`, successfully linking the low-level SPI and GPIO drivers (illustrated in

Figure). Crucially, the sensor hardware remains physically unpowered during this standby phase (the `ENABLE` pin is held low, drawing $< 1 \mu\text{A}$) to conserve battery.

The second stage, the actual radar boot sequence, is strictly on demand. It is triggered when the host App transmits a target mode command (e.g., Vibration) followed by a `START` command via the BLE Control Service. Upon receiving this sequence, the adapter acts as a dynamic dispatcher, invoking the `Radar_Adapter_Start()` function to route execution to the corresponding mode-specific initialization routine (such as `init_vibration()`). This routine loads the user-defined configuration parameters into the SDK context and dynamically allocates RAM for the appropriate sensor buffers. Only at this exact moment is the `ENABLE` pin asserted to physically power the A121 hardware, allowing the SDK to create, calibrate, and prepare the sensor. Once this initialization is complete, the cooperative task scheduler takes over, continuously invoking the `Radar_Adapter_Process()` function to handle radar measurement polling and algorithmic execution. This modular dispatcher architecture ensures that the complex radar pipeline only consumes power and RAM when active monitoring is required, whilst keeping the codebase highly scalable for future algorithmic modes.

3.5.2 Vital signs

The vital signs mode was developed to extract breathing rate and heart rate from small chest-wall movements using the phase information of the A121 mmWave radar signal. Unlike the vibration mode, which monitors a user-defined fixed measurement point, vital-sign monitoring must first identify the range bin that most likely corresponds to the subject's chest and then track weak physiological motion in real time. The final algorithm was therefore developed through an iterative process: an initial distance-based method was first evaluated, its limitations were identified through testing, and the system was then redesigned into a phase-based signal-processing pipeline with range-bin selection, respiratory-interference suppression, spectral estimation, and BPM stabilisation.

Initial distance-based approach and limitations.

At the beginning of the development process, the standard Acconeer distance detector, `acc_detector_distance`, was considered as the first approach. This detector estimates target distance by identifying the strongest reflected signal within discrete range bins. Although this method is suitable for general distance measurement and object detection, it was found to be unreliable for vital-sign extraction. In practical tests, the output distance frequently jumped between neighboring range bins even when the subject was approximately stationary. These jumps were much larger than the actual chest displacement caused by respiration or heartbeat and therefore produced artificial motion that dominated the physiological signal.

The main limitation of this distance-based approach is that the detector is primarily based on amplitude peak selection. It reports the location of the dominant reflection peak rather than preserving the coherent phase variation of a fixed target. As a result, the measurement is sensitive to range-bin switching, multipath reflections, and small posture changes. For vital-sign monitoring, the useful motion is usually at the sub-millimetre level, especially for heartbeat-induced chest-wall displacement. A distance output based only on discrete peak detection does not provide sufficient sensitivity or stability for this task. Even when exponential smoothing was applied, the bin-switching artefacts could not be fully removed because they were structured discontinuities caused by the detector logic rather than random noise. This motivated the transition from distance-based sensing to phase-based sensing.

Transition to phase-based sensing.

The final vital-sign pipeline uses the Acconeer presence detector, `acc_detector_presence`, because it provides access to complex IQ radar data across multiple range bins. For each selected range bin, the radar return can be represented as a complex signal,

$$z(t) = I(t) + jQ(t)$$

where $I(t)$ and $Q(t)$ are the in-phase and quadrature components. The instantaneous phase is then extracted as

$$\phi(t) = \arg(I(t) + jQ(t))$$

Since the radar signal travels from the sensor to the subject and back, a small chest displacement produces a round-trip phase change. The relationship between phase variation and radial displacement is given by

$$\Delta d(t) = \frac{\lambda}{4\pi} \Delta \phi(t)$$

For the A121 operating near 60 GHz, the wavelength is approximately 5 mm, giving a displacement sensitivity of roughly 0.4 mm per radian. This means that even small phase variations can represent meaningful chest-wall motion. This phase-displacement relationship became the foundation of the vital-sign algorithm, allowing the system to detect respiration and heartbeat more accurately than the original distance-based method.

The implementation first averages the complex IQ samples across all sweeps within a frame for the selected range bin (see **Appendix A, Figure A1 for the detailed phase extraction and unwrapping implementation**). The phase is then extracted using `cargf(mean)`. To avoid artificial discontinuities caused by phase wrapping, the phase difference between consecutive frames is corrected whenever it exceeds the $\pm\pi$ boundary. The corrected difference is accumulated to reconstruct a continuous unwrapped phase signal for subsequent physiological analysis.

Range-bin selection using a coarse sweep.

A key challenge in phase-based vital-sign sensing is selecting the correct range bin. Simply choosing the nearest bin or the bin with the strongest reflection is not always reliable, because the radar may receive strong reflections from walls, furniture, clothing, or other body parts. Therefore, the implemented system uses a coarse sweep before entering the fine measurement stage.

During the coarse sweep, the presence detector scans the monitoring region from 0.3 m to 2.5 m at 20 Hz. For each candidate range bin, the system records a short window of phase data and evaluates the spectral signal quality in both the respiration band and the heart-rate band. The respiration band is defined as 0.15–0.70 Hz, while the heart-rate band is defined as 0.90–3.00 Hz. Rather than selecting the strongest amplitude reflection, the algorithm compares the physiological signal quality of candidate bins and selects the closest bin that satisfies the required heart-rate SNR threshold. This reduces the risk of locking onto static clutter such as walls or furniture.

Each candidate bin is evaluated using `compute_coarse_snr()` in the respiration and heart-rate frequency ranges (refer to the **coarse range-bin selection logic provided in Appendix A, Figure A2**). A bin is accepted only when the heart-rate SNR is above the minimum threshold, and the closest valid bin is selected because the bins are ordered from near to far. After the best bin is selected, the stored coarse-sweep data can be used to initialize the fine processing buffers. This hot-start strategy reduces the initial waiting time before stable BPM estimates can be produced.

Initial signal processing and observed problems.

Once the target bin is selected, the raw wrapped phase is converted into a continuous unwrapped phase signal. The first version of the signal-processing pipeline applied simple Butterworth band-pass filters directly to this unwrapped phase. The respiration component was extracted using a 0.15–0.70 Hz band-pass filter, corresponding to approximately 9–42 breaths per minute. The cardiac component was extracted using a 0.90–3.00 Hz band-pass filter, corresponding to approximately 54–180 beats per minute.

This initial pipeline was able to produce reasonable breathing-rate estimates because respiration produces relatively large and slow chest-wall motion. However, the heart-rate estimate was much less stable. The first problem was that respiration is much stronger than heartbeat, and its harmonics can fall inside the cardiac frequency band. For example, the third to sixth harmonics of the breathing frequency may overlap with the expected heart-rate range. As a result, the heart-rate power spectrum can be dominated by respiratory interference rather than the actual heartbeat. The second problem was that the high-frequency cardiac component often had a lower signal-to-noise ratio, making the FFT

peak less clear. The third problem was output instability: the estimated BPM values could change abruptly from one update to the next, which made the mobile display difficult to interpret.

Even after separating the nominal respiration and heart-rate bands, the implementation still needed to explicitly suppress breathing-related harmonic bins in the heart-rate power spectrum (**the specific harmonic suppression logic is detailed in Appendix A, Figure A3**). This confirmed that respiratory leakage was a major source of error in the initial heart-rate estimation pipeline.

Refined processing pipeline with drift removal and adaptive cancellation.

To improve robustness, the final implementation introduces several additional processing stages. First, a 0.05 Hz high-pass filter is applied to the unwrapped phase signal to remove DC offset and slow baseline drift. This prevents long-term body-position changes and accumulated phase bias from affecting the band-pass filters. The respiration signal is then isolated using the 0.15–0.70 Hz Butterworth band-pass filter.

For the heart-rate path, an adaptive noise cancellation stage is inserted before the cardiac band-pass filter. The system uses a 16-tap LMS adaptive noise canceller with a delay line. The purpose of this stage is to estimate the respiration-correlated component in the high-pass-filtered phase signal and subtract it before extracting the heartbeat. The LMS step size is set to 0.002, and a normalised update step is used to improve convergence stability. A leaky term is also applied to prevent long-term drift in adaptive weights. This design reduces the influence of respiratory leakage and harmonics before the signal enters the cardiac band-pass filter.

The LMS stage estimates the periodic breathing component from delayed reference samples and subtracts it from the high-pass-filtered phase signal (**the complete C implementation of this adaptive noise cancellation logic is provided in Appendix A, Figure A4**). The resulting cleaned signal is then passed into the heart-rate Butterworth filter. This refinement directly addresses the main limitation of the initial pipeline, where respiratory harmonics could dominate the cardiac frequency band.

After filtering, the algorithm stores the breathing and heart-related signals in circular buffers with a 128-sample window. A windowed FFT is then performed periodically, with zero-padding used to improve spectral sampling. Gaussian peak interpolation is applied around the dominant spectral bin to obtain sub-bin frequency accuracy. The power spectrum is also smoothed across updates, which reduces the effect of single-frame noise and makes the estimated rates more stable.

Output stabilisation and long-term robustness. The final breathing and heart frequencies are converted into breaths per minute and beats per minute. To reduce sudden jumps in the displayed results, a five-sample median filter is applied to the estimated BPM outputs. This filter removes isolated outliers while preserving slower physiological changes. Since only five recent valid estimates are sorted, the computational cost is negligible for the STM32 edge platform.

To stabilize the final output, a lightweight five-sample median filter is applied to the estimated BPM (**as detailed in the firmware code snippet in Appendix A, Figure A5**). This stage was introduced because the raw FFT-based estimate could occasionally jump due to transient noise, weak cardiac SNR, or temporary spectral ambiguity. The median filter improves the user-facing stability of both breathing-rate and heart-rate outputs without requiring a heavy computational model.

The algorithm also reports a smoothed target distance, calculated using an exponential moving average of the selected range-bin distance. The final vital sign output is transmitted to the mobile application through the custom BLE Vital Sign Service, with each notification containing breathing rate, heart rate, and target distance. The vital-sign processing routine is only activated after the system has completed target searching and coarse-bin selection, preventing unreliable early estimates from being displayed.

Because phase unwrapping accumulates phase changes over time, the system also includes a phase-renormalisation mechanism. Approximately every 600 seconds, the vital-sign processing state is reinitialised to prevent floating-point precision degradation and long-term phase drift. If the target is lost for an extended period, the system resets the selected bin and returns to the searching stage. This

allows the algorithm to recover when the subject moves away, changes posture significantly or leaves the monitoring area.

Overall, the final vital-sign algorithm is not a direct use of the radar SDK output. It combines IQ-based phase extraction, phase unwrapping, coarse range-bin selection, DC drift removal, adaptive respiratory-interference cancellation, FFT-based spectral estimation, Gaussian peak interpolation, and median-filtered BPM stabilisation. Compared with the initial distance-based approach, this iterative design significantly improves the robustness of both breathing-rate and heart-rate estimation under realistic indoor sensing conditions.

3.5.3 Vibration

The vibration monitoring mode measures sub-millimeter target displacement and dominant oscillation frequencies at a configurable distance. The processing chain is deeply integrated with the Acconeer RSS SDK, operating on a three-level signal hierarchy: individual radar pulses (sweeps), hardware-averaged sweeps, and aggregated time-series frames.

Complex IQ Phase Extraction and Unwrapping

The conversion from raw radar reflections to physical displacement relies on coherent phase extraction. Operating at the user-defined `measured_point` (range bin), the `acc_processing_execute` SDK function processes the raw ADC buffer to yield a complex In-phase and Quadrature (IQ) sample. The algorithm extracts the phase angle $\phi = \arctan(Q/I)$ to encode the round-trip distance. Because the carrier wavelength $\lambda \approx 5 \text{ mm}$ for the 60.5 GHz operating frequency, the relationship between phase shift and physical movement is governed by $\Delta\text{phase} = \frac{4\pi \times \Delta\text{distance}}{\lambda}$. Consequently, a $1 \text{ }\mu\text{m}$ target displacement produces a minute phase shift of approximately 0.0025 rad.

Because the raw phase is mathematically bounded to the interval $[-\pi, \pi]$, physical movements exceeding $\lambda/4$ (approximately 1.25 mm) between sweeps create artificial phase wrap-around discontinuities. The SDK's `acc_vibration_process` module actively unwraps the phase to reconstruct the true continuous displacement waveform. Following unwrapping, a low-pass filter is applied to the time-series data to aggressively reject high-frequency electronic noise. Depending on the selected operational profile, the firmware dynamically configures the filter coefficient to 0.5 (for the High Frequency preset) or 0.8 (for the Low Frequency preset).

Spectral Analysis via Fast Fourier Transform (FFT)

Once the phase time-series is filtered, the SDK executes a 1024-point Fast Fourier Transform (FFT) to convert the displacement data into the frequency domain. The frequency resolution of this resulting spectrum is directly defined by the ratio of the radar's sweep rate to the time-series length.

The system provides two distinct operational presets to adapt to different frequency bands:

- **High Frequency Preset:** Aggregates 1024 sweeps per frame at a 10,000 Hz sweep rate. This configuration yields a frequency resolution of roughly 9.8 Hz per bin, allowing the system to detect high-speed oscillations up to a Nyquist limit of 5,000 Hz.
- **Low Frequency Preset:** Operates at a 200 Hz sweep rate, bounded to a 100 Hz Nyquist limit, achieving an ultra-fine frequency resolution of approximately 0.2 Hz per bin. The algorithm scans the FFT magnitude spectrum to identify dominant peaks, dynamically converting these specific bins back into peak physical displacement values (measured in micrometers).

Custom Stability Gating and BLE Transmission

While the Acconeer RSS SDK successfully extracts raw FFT peaks, directly broadcasting these values over the BLE link would result in high false-positive rates due to transient environmental noise. To ensure absolute data reliability before it reaches the user interface, a custom application-level post-processing gate was engineered within the `radar_adapter.c` abstraction layer.

For each processed frame, the custom firmware layer compares the newly detected dominant frequency ($f_{current}$) against the previously recorded frequency (f_{prev}). If the frequency difference is strictly $< 0.5\text{Hz}$, an internal `stability_counter` is incremented. The system only validates the signal when two strict conditions are simultaneously met: the frequency must remain stable for at least three consecutive frames (`stability_counter` ≥ 3), and the measured displacement amplitude must exceed $5\mu\text{m}$.

Upon satisfying this stability gate, the firmware packages the verified frequency and peak displacement metrics into the BLE notification payload, transmitting them to the iOS application via the `FE72` characteristic. Conversely, if the target signal decays and fails the stability gate, the firmware systematically transmits a zero-value packet to clear the application's dashboard, ensuring a clean and artifact-free user experience.

3.6 Mechanical Design

3.6.1 Lens Design Details

The design of this mechanical structure is not solely for providing chassis protection; On the contrary, the geometric shape of the chassis directly affects the electromagnetic focusing characteristics of the radar system. By setting the distance between the sensor and the lens to achieve optimal performance.

To improve the sensing capability of the mmWave radar system, a 3D-printed enclosure integrated with a dielectric lens was designed and fabricated. The primary purpose of the lens was to increase antenna directivity and enhance the received signal strength for physiological monitoring applications.

According to Acconeer lens design guides (Acconeer, 2024), dielectric lenses are a cost-effective solution for shaping mmWave radiation patterns and improving antenna gain. The lens focuses the emitted electromagnetic waves into a narrower beam, thereby increasing signal intensity and reducing unwanted reflections and environmental clutter. This is particularly beneficial for heartbeat and respiration monitoring, where weak chest-wall motion must be extracted from low-amplitude radar reflections.

The lens geometry is modeled using Acconeer's official website and manufactured using a 3D printer made of thermoplastic materials. We use 3D printing technology to rapidly prototype the radar housing and lens structure. Many thermoplastic materials have low dielectric loss and low dielectric constant, making them suitable for the manufacture of millimeter wave lenses. In addition, 3D printing enables rapid iteration of lens geometry, simplifying the integration between radar modules and casings.

The lens used is a Fresnel Zone Plate (FZP) lens. The structure of FZP is a stepped structure composed of dielectric rings of different thicknesses. Unlike refractive lens such as plano-convex lens, the annular thickness (and total thickness) of FZP lens is independent of the radius of the lens. This feature allows for the manufacture of large-diameter lenses without increasing thickness. In contrast, refractive lens typically require a higher thickness to maintain optical performance. Meanwhile, FZP also reduces injection molding risks. Manufacturing thick lenses is challenging and prone to sink marks or other deformations. The thin design of FZP can effectively avoid these process limitations.

Although in actual engineering, the dielectric constant deviation of the material and the standing wave effect of the enclosure radome have an impact on the final gain, fine-tuning is required, but the initial geometric dimensions and spatial relative distances must be determined through rigorous theoretical calculations.



Figure 12: FZP Lens 3D Model

The A121 pulse coherent radar operates in the millimeter-wave band with a center frequency of 60 GHz. So the Free-Space Wavelength (λ_0) is:

$$\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{60 \times 10^9 \text{ Hz}} = 5 \text{ mm}$$

This lens design intends to use 3D printing technology to process FZP lenses, and the consumables used are solid coloured polylactic acid. Although the theoretical dielectric constant of solid polylactic acid (PLA) raw material is about 2.7 to 2.8, considering the composite effect of micro interlayer air gaps on electromagnetic wave propagation in 3D printing technology, the equivalent relative dielectric constant of its finished product will slightly decrease. Therefore, in this design, the equivalent relative dielectric constant of PLA is conservatively set to $\epsilon_r = 2.6$. We don't need very precise values because we still need to manually finetune during physical testing, so we only need to estimate approximate values.

The Wavelength in Medium (λ) is:

$$\lambda = \frac{\lambda_0}{\sqrt{\epsilon_r}} = \frac{5 \text{ mm}}{\sqrt{2.6}} \approx 3.1 \text{ mm}$$

To balance process complexity and focusing efficiency, a quarter-wavelength phase correction is chosen, $P = 4$ (total of 1 main zone, containing 4 stepped sub-zones, with $N = 4$ sub-zone layers). Here, the focusing distance from the sensor chip to the lens reference plane is set as $F = 6 \text{ mm}$.

Afterwards, key parameters need to be calculated:

According to the phase-corrected FZP lens design theory, the calculation formulas for the tangential radius r_i of each stepped ring and the step height s are as follows:

Calculation of Each Stepped Zone Radius The inner and outer diameter calculation formula for each annular zone is:

$$r_i = \sqrt{2Fi \frac{\lambda_0}{P} + \left(\frac{i\lambda_0}{P}\right)^2} \quad (i = 1, 2, 3, \dots, N)$$

Substituting $F = 6 \text{ mm}$, $\lambda_0 = 5 \text{ mm}$, $P = 4$, $N = 4$ into the formula:

When $i = 1$ (inner diameter of the first-level step):

$$r_1 = \sqrt{2 \times 3.5 \times 1 \times \frac{5}{4} + \left(\frac{1 \times 5}{4}\right)^2} \approx 4.07 \text{ mm}$$

When $i = 2$::

$$r_2 = \sqrt{2 \times 6 \times 2 \times \frac{5}{4} + \left(\frac{2 \times 5}{4}\right)^2} \approx 6.02 \text{ mm}$$

When $i = 3$::

$$r_3 = \sqrt{2 \times 6 \times 3 \times \frac{5}{4} + \left(\frac{3 \times 5}{4}\right)^2} \approx 7.68 \text{ mm}$$

When $i = 4$ (outermost radius of the lens):

$$r_4 = \sqrt{2 \times 6 \times 4 \times \frac{5}{4} + \left(\frac{4 \times 5}{4}\right)^2} \approx 9.21 \text{ mm}$$

From this, the total design diameter of the FZP lens is $D = 2 \times r_4 \approx 18.44 \text{ mm}$. At this point, the focal length to diameter ratio $F/D = 6/18.44 \approx 0.325$.

It is worth noting that to balance material thickness, injection modelling process defects, and sidelobe suppression, the focal diameter ratio F/D of traditional refractive lenses (such as plano-convex and convex-planar lenses) must be limited between 0.3 to 0.6. The FZP lens used in this design significantly increases its tolerance for F/D . Due to the diffraction principle, the overall structural thickness of FZP lenses is determined only by the phase delay order (P) and the dielectric constant of the material and is independent of the lens aperture or focal length. This physical property gives the housing design a high degree of flexibility: even in flat housing applications that require extremely compact structures and ultra short focal lengths ($F/D < 0.3$), FZP lenses will never have process problems such as thick edges or injection shrinkage marks.

The parameters we have adopted are currently within the range, but there is still a lot of room for adjustment without worrying about exceeding the range and causing lens failure.

To achieve a $\frac{\lambda_0}{4}$ phase delay, the theoretical thickness s of a single-layer annular step is calculated as:

$$s = \frac{\lambda}{P(\sqrt{\epsilon_r} - 1)} = \frac{3.1}{4 \times (\sqrt{2.6} - 1)} = \frac{3.1}{4 \times (1.6124 - 1)} \approx 2.04 \text{ mm}$$

The total thickness of the 4-layer steps at the center of the lens is:

$$T = N \times s = 4 \times 2.04 \text{ mm} \approx 8.16 \text{ mm}$$

After calculating the geometric dimensions of the lens itself, the theoretical placement distance of the sensor inside the enclosure (spatial air span from the lens bottom surface to the A121 chip surface) needs to be corrected in combination with the electromagnetic boundary of the radome:

1. Theoretical Air Focal Length Reference (F_{air}): The focal length $F = 6 \text{ mm}$ obtained from basic ray optics calculation. However, due to the phase velocity change when high-frequency electromagnetic waves penetrate the solid thickness of the FZP lens, and the fact that the total solid thickness ($T \approx 8.1 \text{ mm}$) is independent of the lens radius, this provides spatial convenience for large-diameter flattening designs.
2. Standing Wave Effect and Periodic Maximum Correction: Full-wave electromagnetic simulation and experimental data show that inside high-frequency enclosures for millimeter-wave radars, complex interference is formed due to reflections from the radome and mechanical environment. Its one-way gain exhibits periodic strong-weak alternating fluctuations with a period of $\lambda_0/2 = 2.5 \text{ mm}$.

Experiments show that under the overall design framework of $F = 6 \text{ mm}$, the actual air gap maximum gain peak point often converges inward, landing at 3.5 mm .

In summary, the final design of the structural stacking parameters for the internal space of the enclosure is as follows:

Lens Design Outer Diameter D : 18.44 mm

Lens Step Total Thickness T : 8.16 mm

Air Layer Spacing from Chip to Lens: The initial enclosure internal slot positioning reference is set to 3.5 mm . In physical testing, the focal length can also be successfully increased.

3.6.2 Electromagnetic Material Trade-Off Analysis

In the design of lenses, the electromagnetic properties of materials, mechanical strength, and microstructure of processing techniques are the core elements that determine the final detection performance of the system. Based on the development cycle and practical application scenarios of this project, the design scheme adopts completely different but complementary material and process optimization strategies for different functions.

According to the Acconeer Lens design guidelines, the material of the lens is Polyamide 12 (PA12). PA12 exhibits excellent electromagnetic transmission performance in the 60 GHz millimeter wave frequency band. Its relative dielectric constant is stable at around 2.6, and the loss tangent value is extremely low (usually less than 0.005), with extremely superior high-frequency transmittance. The absorption and thermal losses when millimeter waves penetrate PA12 lenses can be almost negligible. The lens is manufactured using an integrated high-pressure injection molding process to create a PA12 shell top cover, which achieves a perfect solid state of 100. From the perspective of physical and mechanical properties, PA12 has excellent toughness, resistance to physical collisions, and wear resistance. Meanwhile, the material has an extremely low moisture absorption rate and excellent resistance to chemical corrosion and UV aging. Whether in changing outdoor climates, humid environments, or facing the self-heating of radar chips, PA12 can ensure the long-term absolute stability of the lens ring geometry, ensuring that the electromagnetic focusing performance of the radar system does not degrade throughout its entire lifecycle.

However, this lens is not universally applicable in all scenarios during actual use. In short distance and wide range testing, the lens has excellent waveform tightening performance, making it necessary to face the measured object directly for testing. This is exactly the opposite of the scenario we envisioned for measuring heartbeat/breathing. Therefore, it is necessary to use other lenses.

Afterwards, we selected solid coloured polylactic acid (PLA) consumables and processed the shell and Fresnel zone plate (FZP) lens using Fused Deposition Modelling (FDM) 3D printing technology.

From the perspective of electromagnetic properties, although the theoretical relative dielectric constant of pure solid PLA raw materials in the microwave frequency range is usually distributed between 2.7 and 2.9, in the actual FDM printing process, extremely small amounts of micro air gaps are inevitably mixed between wires and layers (the relative dielectric constant of air is strictly 1.0). This composite physical structure of matrix plastic and trace air will lead to the overall inward convergence of the equivalent relative dielectric constant of printed products in the 60 GHz frequency band. Engineering experience and parameter calibration have shown that conservatively setting the equivalent relative dielectric constant of PLA printed products to 2.6 not only conforms to the real physical state after 3D printing processing but also allows for direct application of standard theoretical formulas for initial geometric calculations of the ring radius and step thickness in each zone.

In specific "short distance, large range" testing scenarios, the introduction of PLA materials and FDM technology has demonstrated unique engineering advantages: Firstly, PLA has a relatively high loss tangent value (usually in the range of 0.01 to 0.02) in the extremely high frequency band of 60 GHz, and its high-frequency dielectric loss is several times that of the officially recommended material. Millimeter waves experience mild energy attenuation when penetrating PLA media, which acts as a natural physical attenuator in radar chip close range detection, effectively reducing echo energy and preventing large signal saturation and baseband signal clipping distortion in radar receivers, thereby making near-field detection data more stable and linear. Secondly, the inherent microscopic interlayer non-uniformity and surface roughness of FDM printing process can cause a certain degree of microscopic diffraction scattering of millimeter waves when passing through the lens. This electromagnetic non-uniformity breaks the extremely strict parallel focusing condition, causing a certain degree of broadening of the originally narrow main lobe beam in theory. On the premise of

allowing a small loss of peak gain, this phenomenon expands the lateral field of view and spatial coverage range of the radar, perfectly meeting the special scenario requirements of short distance and relatively large-scale detection (such as heart rate detection).

However, PLA also has significant physical limitations. Its material texture is relatively brittle, with weak impact resistance, and its glass transition temperature is only around 60 degrees Celsius. PLA structures are prone to thermal creep and geometric deformation in environments with long-term high load heating of radar chips or outdoor high temperature exposure in summer. Once the annular step size of the FZP lens undergoes micrometer level distortion, its phase correction mechanism will completely fail, resulting in the radar being unable to focus accurately. Therefore, PLA is only used as the optimal solution for prototype testing and functional verification stages.

Therefore, in different usage scenarios, we will use different lenses for testing. These observations highlight the engineering trade-off between electromagnetic efficiency, mechanical robustness, and sensing coverage when selecting lens materials for compact millimeter-wave physiological monitoring systems.

3.6.3 Electromagnetic-Mechanical Integration

The enclosure geometry was developed using parametric CAD modelling to ensure precise alignment between the radar module, dielectric lens, and internal mounting structures. Parametric constraints enabled rapid iteration of the lens-to-sensor spacing during experimental optimisation while maintaining dimensional consistency across repeated design revisions.

The use of parametric modeling enables rapid iterative optimization of shell geometry in experimental testing. Especially, the installation height of the lens, internal air gap spacing, and PCB alignment can be adjusted while maintaining overall dimensional consistency. This ability is important because small geometric deviations can significantly affect beam focusing characteristics and sensing performance when operating at 60 GHz.

In addition, importing the STEP model of the completed PCB into the CAD environment allows for checking for interference between the casing and assembled components before manufacturing. This reduces mechanical integration errors and accelerates the prototyping process of repeatedly replacing lenses and casings.

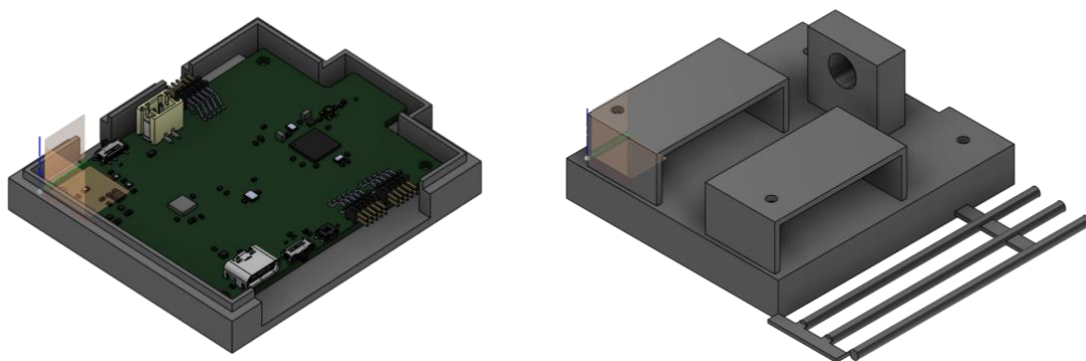


Figure 13: Bottom of the shell with PCB Board 3D Model (left) and Top of the 3D Model (right)

The main contribution of this work is the integration of low-cost 3D printed media lenses into portable millimeter wave medical monitoring platforms. Unlike traditional naked radar configurations, this design improves signal directionality and increases received signal strength through passive

electromagnetic focusing. In addition, the customized casing achieves stable positioning between the radar sensor and the lens, while maintaining a compact system size suitable for portable healthcare applications. The use of fast 3D printing prototypes also reduces manufacturing costs and allows for iterative optimization of lens structures to achieve physiological sensing performance.

3.6.4 Assembled product

The following shows the soldered PCB board and the assembled finished product.

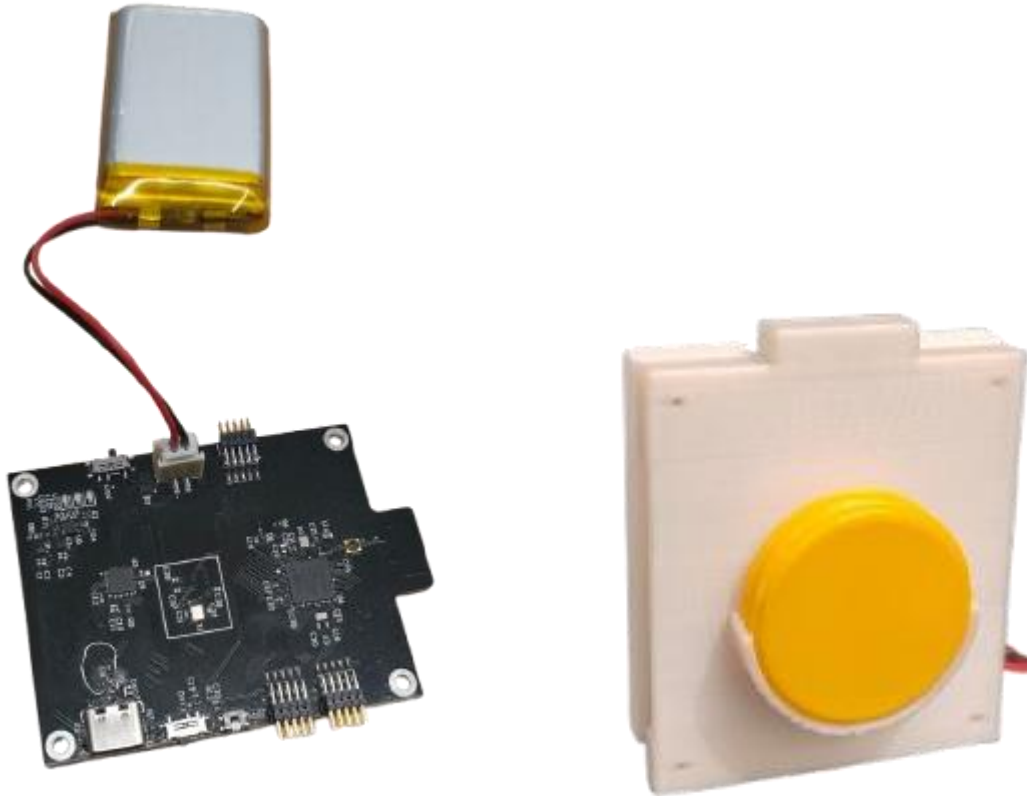


Figure 14 (Left): Completed PCB board after soldering.

Figure 15 (Right): Finished product assembled with 3D printed shell and lens.

3.7 Result and Analysis

3.7.1 Vibration Frequencies

We conducted vibration testing with a distance of 0.2m between the sensor and a tuning fork that can emit a fixed frequency, tapping the tuning fork produced a frequency, and the sensor received data.

Table 2: Fixed distance tuning fork vibration detection

Standard (Hz)	512	480	426.6	384	341.3	320	288	256
Measured (Hz)	517.58	488.28	429.69	390.62	341.8	322.27	292.97	253.91
Difference	5.58, 1.09%	8.28, 1.73%	3.09, 0.72%	6.62, 1.72%	0.50, 0.15%	2.27, 0.71%	4.97, 1.73%	-2.09, 0.82%

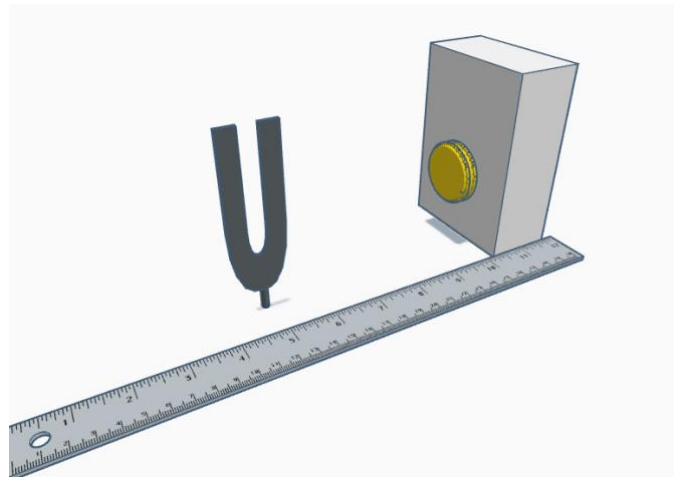


Figure 16: Schematic picture of tuning fork test, the yellow one is the sensor with lens

The vibration detection algorithm performs well overall and has the following main advantages:

High measurement accuracy: The maximum relative error within the testing range of 256-512 Hz is 1.73%, which meets the application requirements of most industrial vibration monitoring.

Good frequency coverage capability: Thanks to the high sweep frequency of 10 kHz, the algorithm can reliably detect mid to high frequency vibrations, with a Nyquist frequency of up to 5 kHz, far exceeding the maximum frequency of this test (512 Hz).

Although the performance is acceptable, there are still the following main shortcomings:

Limited frequency resolution: The 1024-point FFT resulted in a frequency resolution of only about 9.77 Hz, which was the main source of error (2-8 Hz) in this test, causing the peak frequency to easily shift towards the nearest bin, resulting in a systematic positive bias.

Spectrum leakage issue: If the window function is not used in the code, peak shift and amplitude attenuation are prone to occur when the true vibration frequency is not located at the center of the FFT bin.

Error volatility: The distribution of errors at different frequency points is uneven, with the maximum error around 480 Hz, and stability needs to be improved. 480 Hz is only 1.48 Hz away from the lower Bin (478.52), but 8.28 Hz away from the upper Bin. Due to spectral leakage, the peak energy is pulled to Bin 50, which results in a significant positive error of 8.28 Hz.

3.7.2 Lens Improvement

To quantitatively evaluate the distance measurement and sensitivity enhancement provided by 3D printed media lenses, a controlled high-frequency vibration measurement alignment was established.

The experiment used a 341.3Hz tuning fork for striking, as 341.3Hz had strong test results in previous tests, reducing interference from our subsequent testing distance. The tuning fork is rigidly clamped at a fixed height and perfectly aligned with the line-of-sight direction of the radar sensor. In the experiment, a mechanical striker is used to consistently excite the tuning fork to ensure consistency in the initial vibration amplitude during different testing cycles.

The experimental results are as follows:

Table 3: Fixed frequency tuning fork vibration detection

Distance(m)	0.2	0.3	0.4	0.5	0.6
Without Lens (Hz)	341.8	341.8	0.0	0.0	0.0

With Lens(Hz)	341.8	341.8	341.8	341.8	341.8
---------------	-------	-------	-------	-------	-------

FFT frequency resolution limitation:

$$\text{Resolution} = 10000 \text{ Hz}/1024 \approx 9.7656 \text{ Hz}$$

This means that the algorithm can only output specific discrete frequency values. Therefore, all output results are 341.8.

The experimental results show that the detection range of the system has been greatly improved after integrating passive electromagnetic lenses. Bare radar sensors can only reliably extract small vibrations of tuning forks within a distance of 0.3 meters, while the platform with integrated lenses successfully captured the same oscillation waveform at a distance of 0.6 meters, doubling the maximum detection distance directly (by 100%).

3.7.3 Real-time vital-sign output demonstration

In order to quantitatively evaluate the performance of the proposed phase based vital sign sensing process, a control experiment was conducted using healthy adult subjects located approximately 0.6 meters away from the radar sensor. Simultaneously use a commercial fingertip pulse oximeter to calculate heart rate and respiration. The final vital sign monitoring pipeline has been successfully integrated with the mobile application interface. As shown in Figure 16, the radar system can display heart rate, respiratory rate, and target distance in real time. The radar system continuously monitors the target for 120 seconds under static indoor conditions. The results are shown in Table 4.

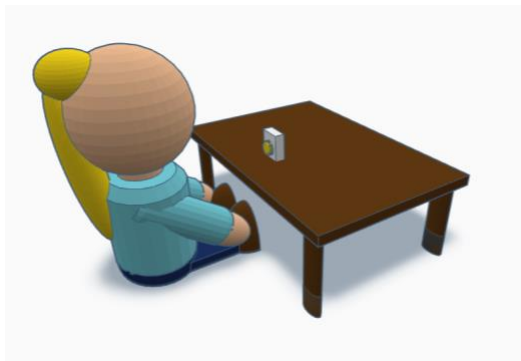


Figure 17 (Left): Schematic picture of real-time vital-sign detection.

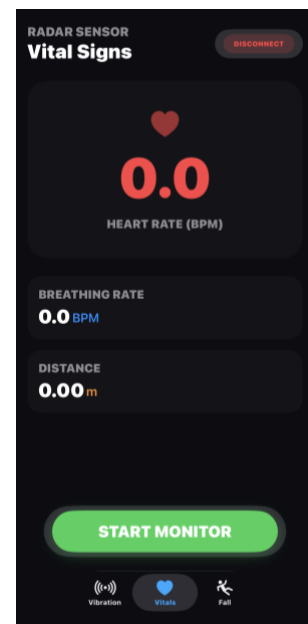


Figure 18 (Right): Real-time vital-sign UI design on the mobile application, showing heart rate, breathing rate, and target distance.

Table 4: Comparison of the proposed system and reference

Measurement	Reference	Proposed System	Absolute Error
Heart Rate	74 BPM	78.1 BPM	4.1 BPM
Breathing Rate	16 BPM	17.2 BPM	1.2 BPM

The results indicate that the proposed phase-based processing pipeline can extract signals with physiological significance using low-power embedded millimeter wave hardware. The estimation error of respiratory rate is relatively low because respiration produces significant periodic chest displacement and has strong spectral advantages. In contrast, heart rate estimation shows a slightly higher deviation due to the significantly weaker amplitude of chest motion induced by the heart and the presence of residual respiratory harmonics. Nevertheless, the successful display of breathing rate, heart rate, and distance on the mobile interface confirms that the embedded radar processing, BLE transmission, and application-level visualization were functionally integrated for vital-sign monitoring.

4 Discussion

4.1 Significance of Results within the Context of Existing Literature

The final system demonstrates that phase-based mmWave sensing provides substantially greater robustness for physiological monitoring compared with conventional distance-based radar measurements. During early development, the use of amplitude-dominant distance estimation resulted in unstable range-bin switching. Although the subject remained stationary, abrupt transitions between adjacent bins introduced artificial displacement variations, confirming that distance estimation is fundamentally insufficient for sub-millimeter vital-sign sensing.

Our novel contribution to the digital signal processing pipeline directly addresses these gaps identified in existing literature. First, the transition to IQ phase-based sensing—coupled with a coarse range-bin selection strategy—avoided static clutter regions and prioritized bins with strong physiological spectral characteristics. This adaptive selection reduced the probability of incorrect target locking. Second, initial testing showed that simple Butterworth band-pass filtering was insufficient because respiratory harmonics frequently overlapped with the cardiac frequency band. By integrating a custom 16-tap Least Mean Squares (LMS) adaptive noise canceller, the system successfully removed respiration-correlated components before cardiac spectral extraction. As a result, we achieved a stable heart rate reading of 78.1 BPM that closely aligned with the 74 BPM reference from a commercial oximeter, highlighting that practical physiological radar sensing requires adaptive interference mitigation alongside traditional filtering.

Furthermore, another **novel contribution** is the integration of low-cost 3D printed Fresnel Zone Plate (FZP) dielectric lenses into a portable millimeter-wave platform. Our empirical results demonstrate that the FDM-printed PLA FZP lens effectively increased the reliable vibration detection range from 0.3 m to 0.6 m, achieving a 100% improvement in spatial coverage while maintaining a compact form factor.

4.2 System Limitations

Despite the promising results, several technical limitations remain. First, the system performance is still sensitive to subject movement, body orientation, and indoor multipath reflections, all of which can degrade phase continuity and reduce the signal-to-noise ratio. Second, the spectral analysis in the vibration monitoring mode is constrained by frequency resolution limits. Utilizing a 1024-point Fast Fourier Transform (FFT) at a 10,000 Hz sweep rate yields a theoretical resolution of approximately 9.77 Hz per bin; this finite sweep window and sampling rate trade-off introduced spectral leakage, which was the primary source of the 8.28 Hz absolute error observed during the tuning fork test. Finally, the 3D-printed PLA lens introduces potential long-term thermal stability concerns under

elevated environmental temperatures, as thermal creep could compromise the electromagnetic focusing mechanism.

5 Future Improvement

5.1 Fall detection: application integration and validation

A firmware-level fall detection prototype was designed and partially integrated into the radar sensing platform. The implemented logic uses a multi-stage state machine to monitor sudden motion changes and sustained displacement from the pre-impact position. The fall state is encoded as a status byte in the BLE notification payload, allowing the firmware to distinguish between normal monitoring, impact-like motion, suspected fall events, and alarm conditions.

However, the fall detection function was not fully integrated into the mobile application within the project timeline. Although the firmware-side algorithm and BLE payload format were prepared, the companion app does not currently parse, display, or respond to the fall-status byte. As a result, fall detection has not yet been validated as a complete end-to-end user-facing function. Future work should therefore focus on three aspects: mobile application integration, algorithm validation, and system-level extension.

In the near term, the mobile application should be updated to parse the extended BLE payload and display the real-time fall state on the monitoring dashboard. The interface could show states such as Normal, Impact, Suspected Fall, and Alarm, so that users or caregivers can clearly understand the system status. When an Alarm state is triggered, the app should also generate a local notification to alert the user even when the monitoring screen is not actively open.

In the medium term, the fall detection thresholds need to be validated using controlled experimental data. The current firmware-level thresholds, including impact score, displacement threshold, and confirmation duration, were selected based on the expected fall behaviour and preliminary reasoning rather than a large dataset of real fall events. Future testing should collect staged fall and non-fall motion data across different sensor distances, body orientations, and indoor environments. This would allow the false-positive rate and false-negative rate to be quantified, and the threshold parameters to be tuned more systematically.

Another important improvement is to reduce the influence of range-bin switching on displacement estimation. Since the A121 presence detector operates over discrete range bins, sudden bin changes may introduce artificial distance jumps. These artefacts could be misinterpreted as large body displacement and may affect the reliability of fall confirmation. Future versions could introduce hysteresis or temporal consistency checks into the bin-selection logic, so that the algorithm only accepts a new range bin when the change remains stable over several frames.

In the longer term, fall detection should be integrated with the rest of the healthcare monitoring workflow. Once an Alarm state is confirmed, the mobile application could automatically send an alert to a designated emergency contact through push notification, SMS, or a cloud relay. The system could also switch to vital-sign monitoring immediately after a fall is detected, allowing caregivers to assess whether the subject still shows breathing and heart-rate activity. This integration would make the platform more suitable for elderly-care and smart-health applications, where fall detection alone is less informative than fall detection combined with post-fall physiological monitoring.

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Appendices

Appendix A: Selected Algorithm Code Snippets

```
86 // Extract the mean phasor across all sweeps in a frame, then update running
87 // unwrapped phase for the given bin. If out_amp is non-NULL, phasor magnitude
88 // (coherence indicator) is written there.
89 static float update_phase(const acc_detector_presence_result_t *res,
90                          const acc_detector_presence_metadata_t *meta,
91                          uint16_t spf,
92                          int bin,
93                          float *prev_a,
94                          bool *first,
95                          float *unwrapped,
96                          float *out_amp)
97 {
98     float complex mean = 0.0f + 0.0f * I;
99
100     for (int s = 0; s < spf; s++)
101     {
102         acc_int16_complex_t p = res->processing_result.frame[s * meta->num_points + bin];
103         mean += (float)p.real + (float)p.imag * I;
104     }
105
106     mean /= spf;
107
108     if (out_amp != NULL)
109     {
110         *out_amp = cabsf(mean);
111     }
112
113     float angle = cargf(mean);
114
115     if (*first)
116     {
117         *prev_a = angle;
118         *unwrapped = angle;
119         *first = false;
120     }
121     else
122     {
123         float diff = angle - *prev_a;
124
125         if (diff > M_PI) diff -= 2.0f * M_PI;
126         if (diff < -M_PI) diff += 2.0f * M_PI;
127
128         *unwrapped += diff;
129         *prev_a = angle;
130     }
131
132     return *unwrapped;
133 }
```

Figure A1: Phase extraction and unwrapping from complex IQ radar samples.

```
int vital_signs_coarse_pick_best(int n_candidates)
{
    coarse_active = false;

    if (n_candidates > COARSE_MAX_CANDS)
    {
        n_candidates = COARSE_MAX_CANDS;
    }

    int closest_slot = -1;

    for (int s = 0; s < n_candidates; s++)
    {
        float r_snr = compute_coarse_snr(s, 0.15f, 0.7f);
        float h_snr = compute_coarse_snr(s, 0.9f, 3.0f);

        // Only log bins that show some signal to avoid flooding the console
        if (h_snr > 1.2f || r_snr > 1.2f)
        {
            printf("[Coarse] Bin %d: resp SNR-%.1f heart SNR-%.1f\n", s, r_snr, h_snr);
        }

        // Accept a bin if the heartbeat SNR is above the minimum threshold
        if (h_snr > 1.5f && closest_slot == -1)
        {
            closest_slot = s; // Take the closest valid bin (bins are ordered near-to-far)
        }
    }

    if (closest_slot != -1)
    {
        printf("[Coarse] Selected closest valid bin: %d\n", closest_slot);
        return closest_slot;
    }

    return -1;
}
```

Figure A2: Coarse range-bin selection based on respiration and heart-rate SNR.

```

712 // --- Heart-rate estimation (0.9-3.0 Hz, 54-180 BPM) ---
713
714 // Suppress breathing harmonics (3rd-6th order) from the heart PSD
715 if (b_ok)
716 {
717     for (int m = 3; m <= 6; m++)
718     {
719         float f_h = m * freq_b;
720         int h_idx = (int)(f_h / freq_delta + 0.5f);
721
722         // Suppress the harmonic bin and its two neighbours
723         for (int offset = -1; offset <= 1; offset++)
724         {
725             int h_min = (int)(0.9f * FFT_N / SAMPLE_RATE_HZ);
726             int h_max = (int)(3.0f * FFT_N / SAMPLE_RATE_HZ);
727             int idx = h_idx + offset;
728
729             if (idx >= h_min && idx <= h_max)
730             {
731                 s_psd_h_smooth[idx] *= 0.15f;
732             }
733         }
734     }
735 }

```

Figure A3: Respiratory harmonic suppression in the heart-rate power spectrum.

```

417
418 // 3. LMS adaptive noise canceller - remove periodic breathing from the heart signal
419 // Shift the reference delay line
420 for (int i = LMS_ORDER + LMS_DELAY - 1; i > 0; i--)
421 {
422     lms_x[i] = lms_x[i - 1];
423 }
424
425 lms_x[0] = hp_diff;
426
427 // Estimate the periodic breathing component
428 float lms_y = 0.0f;
429
430 for (int i = 0; i < LMS_ORDER; i++)
431 {
432     lms_y += lms_w[i] * lms_x[i + LMS_DELAY];
433 }
434
435 // Cancel the breathing estimate from the input signal
436 float raw_clean = hp_diff - lms_y;
437
438 // Compute reference input energy for normalised step size
439 float lms_energy = 1e-3f;
440
441 for (int i = 0; i < LMS_ORDER; i++)
442 {
443     float val = lms_x[i + LMS_DELAY];
444     lms_energy += val * val;
445 }
446
447 float norm_step = lms_mu / lms_energy;
448
449 if (norm_step > 0.05f)
450 {
451     norm_step = 0.05f; // Hard limit to guarantee convergence during transients
452 }
453
454 // Update LMS weights with leaky LMS to prevent long-term drift
455 for (int i = 0; i < LMS_ORDER; i++)
456 {
457     lms_w[i] = 0.999f * lms_w[i] + 2.0f * norm_step * raw_clean * lms_x[i + LMS_DELAY];
458 }
459
460 // 4. Heart-rate band-pass filter applied to the noise-cancelled signal
461 float clean_h = apply_butterworth_h(raw_clean);

```

Figure A4: LMS adaptive noise cancellation for suppressing respiration-related interference before heart-rate filtering.

```

102 // Insertion-sort median over up to BPM_HIST_N values (n<=5, negligible cost)
103 static float compute_median5(float *buf, int n)
104 {
105     float tmp[BPM_HIST_N];
106
107     for (int i = 0; i < n; i++)
108     {
109         tmp[i] = buf[i];
110     }
111
112     for (int i = 1; i < n; i++)
113     {
114         float v = tmp[i];
115         int j = i - 1;
116
117         while (j >= 0 && tmp[j] > v)
118         {
119             tmp[j + 1] = tmp[j];
120             j--;
121         }
122
123         tmp[j + 1] = v;
124     }
125
126     return tmp[n / 2];
127 }

```

Figure A5: Five-sample median filtering for stabilizing BPM output.

Appendix B: Project Source Code and Technical Documentation

The complete firmware source code, Bluetooth Low Energy (BLE) integration logic, and detailed developer manuals for this Capstone project are hosted on GitHub. The repositories contain the full commit history reflecting the hardware-software co-design process. The technical documentation is organized below by subsystem:

B.1 Main Project Repository

- **STM32WBA64 Firmware Codebase (BLE Radar Integration Branch):**
https://github.com/yuhengli001/test_sensor/tree/BLE_Radar_Integration

B.2 Radar Processing & Firmware Architecture

- **Vibration Mode Technical Reference & Developer Manual:**
https://github.com/yuhengli001/test_sensor/blob/BLE_Radar_Integration/docs/developer_manual.md
- **Radar Code Flow & Execution State Machine:**
https://github.com/yuhengli001/test_sensor/blob/BLE_Radar_Integration/docs/radar_code_flow.md

B.3 Bluetooth Low Energy (BLE) & Application Layer

- **STM32WBA BLE Code Setup & Middleware Guide:**
https://github.com/yuhengli001/test_sensor/blob/BLE_Radar_Integration/docs/stm32_ble_code_setup.md
- **CubeMX BLE Parameters Configuration:**
https://github.com/yuhengli001/test_sensor/blob/BLE_Radar_Integration/docs/cubemx_ble_parameters.md
- **Custom BLE GATT Profile Specification:**
https://github.com/yuhengli001/test_sensor/blob/BLE_Radar_Integration/docs/ble_gatt_profile.md
- **BLE Radar Companion Application Specification:**
https://github.com/yuhengli001/test_sensor/blob/BLE_Radar_Integration/docs/ble_radar_app_spec.md

Appendix C: 3D Printing

- **3D Modeling:**

Autodesk Fusion360 can achieve seamless integration and collaborative design. For completed PCB designs, we can directly import a 3D model (.step file) containing precise component height and position information. This integrated feature allows us to perform precise interference checks in the early stages of modeling, effectively avoiding collisions between the inner walls of the casing and large-sized components. At the same time, Fusion360's parametric modeling also has strong efficiency. In the process of box modeling, Fusion 360's parameterization engine allows us to drive geometric shapes by defining variables. Only a small number of parameter values need to be modified, and the model can be automatically reconstructed. This constraint-based modeling approach significantly reduces modeling time costs. And

compared to direct modeling, parameterized logic is more in line with the requirements of design specifications and tolerance control.

- 3D Printing: <https://cloud.3dprinteros.com/>
- 3D Printing Setting:

The screenshot displays the 'Simple Mode' settings for 3D printing. At the top, there are tabs for 'Simple Mode', 'Extra', and 'JSON'. Below these, the 'Printer Type' is set to 'Ultimaker 2+ Connect' and the 'Profile' is 'PLA (0.4 mm) - Normal'. The 'Print Core' is '0.4 mm' and the 'Material' is 'PLA'. The main area features a 3D preview of a yellow object on a blue grid. To the right of the preview are sliders and input fields for 'Layer Height' (0.2 mm), 'Wall Thickness' (0.8 mm), and 'Infill Density' (20%). Below these are 'Printing Temperature' (200 °C), 'Build Plate Temperature' (60 °C), and 'Build Plate Adhesion Type' (Brim). A 'Generate Support' checkbox is also present. At the bottom, there are 'BACK', 'SLICE', and 'SLICE & TOOLPATH PREVIEW' buttons.

Parameter	Value	Unit
Layer Height	0.2	mm
Wall Thickness	0.8	mm
Infill Density	20	%
Printing Temperature	200	°C
Build Plate Temperature	60	°C
Build Plate Adhesion Type	Brim	

Figure A6: Settings for 3D Printing